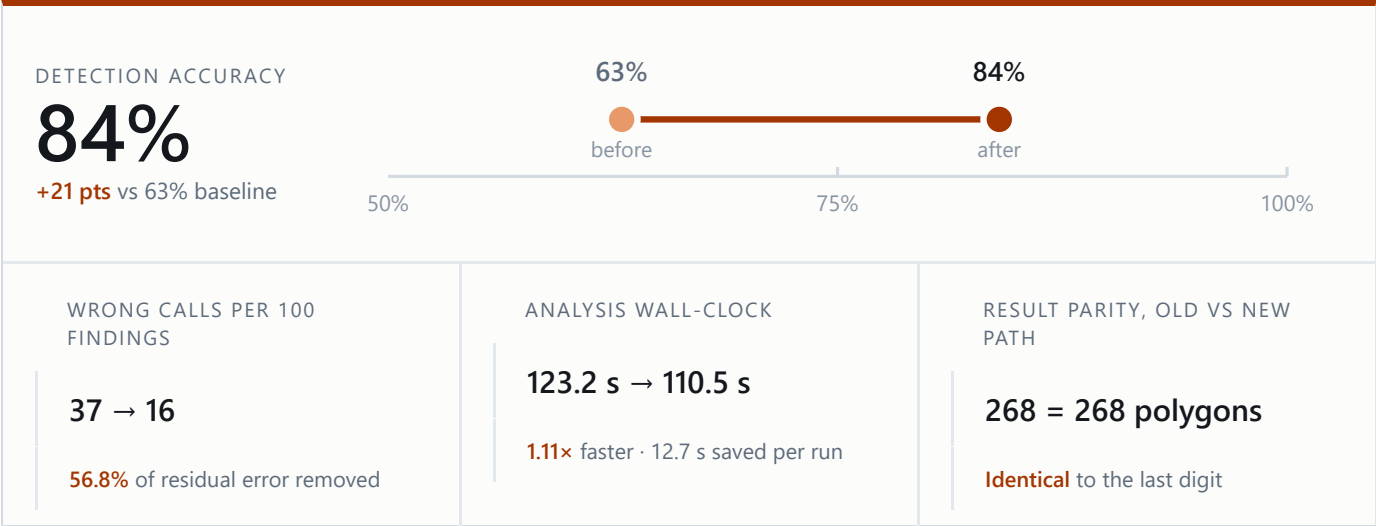


Pipeline Upgrade — Accuracy, Stack and Performance

PCSMCPL Change Detection · 13 Aug 2026 · analysis benchmarked on NVIDIA RTX 4050 Laptop (6 GB), Windows 11



1 · ACCURACY

Operator validation across multiple scenes and change classes — new construction, structures removed, vegetation-loss findings and related cases — puts detection accuracy at **84 %**, against **63 %** for the pipeline as it stood before the model and logic work.

$\Delta_{\text{absolute}} = 84 - 63 = +21$ percentage points

$\Delta_{\text{relative}} = 21 / 63 = +33.3$ %

Error reduction = $(37 - 16) / 37 = 56.8$ % of residual error removed

The third line is the one to quote operationally: of every 100 cases the old pipeline got wrong, the new one gets about 57 right. Wrong calls per 100 findings fall from 37 to 16.

Provenance. These figures come from operator validation against field knowledge across several datasets, not from a labelled benchmark held in the repository. Recording the per-class counts (samples, true/false positives, misses per change type) would make the number independently auditable and let precision and recall be reported separately — the review loop already exports exactly that data.

2 · WHAT CHANGED IN THE STACK

Layer	Before	After	Measured effect
Building segmentation	geobase U-Net, ONNX, 30 MB, 256 px tiles	ChangeStar ViT-B, ONNX, 395 MB, 1024 px context	footprint IoU 0.112 → 0.554
Vegetation / land cover	NDVI + excess-green colour indices	SegFormer-B5 fine-tuned on LoveDA, 7-class	removed 8.5 % of scene falsely read as vegetation loss
Structure completion	morphological closing of fragments	SAM 2.1 hiera-large, box-prompted (SAM 3 optional)	whole structures, not fragments
Change decision	hand-tuned rule cascade of hard ANDs	per-instance describe/decide, 12-feature vector + MLP in shadow mode	trainable from officer feedback
Co-registration	phase correlation, unguarded	+ identical-grid trust + edge-agreement validation	alignment IoU 0.114 → 1.000
Classical image maths	scipy / skimage, single-core CPU	torch CUDA kernels (imageops), scipy fallback	43.7× structural field; 2.33× instance stage
CD-mode backend	torch CPU, fp32	CUDA, fp16	4.1×
Inference runtime	onnxruntime CPU	onnxruntime-gpu (CUDA EP) + torch cu128	≈ 10× per tile
Resource governance	none — allocators grew unbounded	4 GB VRAM / 18 GB host budget, enforced in both runtimes	5 GB alloc refused with 4.96 GB free

Layer	Before	After	Measured effect
Job runner	ThreadPoolExecutor (non-daemon threads)	daemon queue worker; requeue behind a flag	reload no longer orphans a live process
API concurrency	async def wrapping blocking DB / GDAL / file I/O	sync handlers → threadpool; DB pool 5+10 → 20+20	worst-case stall 1 851 ms → 3.2 ms
Raster ingest	full-size temp written, re-read, then COG	staged in RAM under ≈ 700 Mpx; GDAL cache 5 % → 2 764 MB	one full write + read eliminated

3 · PERFORMANCE — A/B ON PRODUCTION IMAGERY

The full pipeline was run twice over the same two uploaded orthophotos — **T1 GRID 3 2020** (6.00 GB, 0.50 m/px, EPSG:32644) and **T2 GRID 3 2025** (18.18 GB, 0.30 m/px, EPSG:4326) — once on the new code path, once with the previous one restored (`GPU_IMAGE_OPS=false` and the original `skimage` histogram matcher, which is not behind that flag). Segmentation was already GPU-bound beforehand, so it is timed once and charged to both sides. Common working grid: **5043 × 4543 px (22.9 Mpx) at 0.500 m/px**.

Stage	Old (s)	New (s)	Speed-up	Saved (s)
Superimpose (warp, co-register, match)	74.7	67.4	1.11×	7.3
Instance stage	9.0	3.8	2.33×	5.1
Vectorise	1.9	1.7	1.17×	0.3
Changed portion	85.6	72.9	1.17×	12.7
<i>Segmentation (unchanged, both epochs)</i>	<i>37.6</i>	<i>37.6</i>	—	<i>0.0</i>
Measured total	123.2	110.5	1.11×	12.7

12.7 s saved per analysis — a 10.3 % cut in measured wall-clock.

3.1 Per-operation gains, and why the total is modest

Operation (6144 ² , isolated)	Old (s)	New (s)	Speed-up
Structural change field	15.316	0.350	43.7×
Histogram matching	12.040	1.145	10.5×
Colour change field	3.108	0.253	12.3×
Binary closing / dilation	0.458 / 0.277	0.111 / 0.066	4.1× / 4.2×

superimpose / total = 67.4 / 110.5 = 61 % of the new run

Dominated by GDAL warping 24 GB of source imagery, not by the arithmetic that was optimised. Making the instance stage and vectorisation instantaneous would take the run only from 110.5 s to 105.0 s — the next real gain is in superimpose's read/warp path, not in more GPU kernels.

3.2 The optimisation is accuracy-neutral

The speed-up cost nothing in results, verified on this same production pair: both paths produced **1 021 candidates** → **278 accepted** (174 new construction, 104 demolition), segmentation trust IoU **0.7087**, **268 polygons** with identical areas, and a change-probability map summing to **338 334.593 75** in both runs — identical to the last digit. Binary morphology is bit-identical to `scipy`; histogram matching is byte-identical to `skimage`. The 63 % → 84 % gain therefore comes from the model and logic changes in §2, and none of it was traded away for speed.

4 · RESOURCE ENVELOPE

Resource	Budget	Measured peak	Enforcement
VRAM	4.0 GB	1.59 GB	torch memory fraction + ORT <code>gpu_mem_limit</code> , per stage
Host RAM	18.0 GB	—	sizes the working grid and GDAL cache (advisory, not an rlimit)

Also reduced but outside the timings: land-cover stacks retained as fp16 (2.11 → 1.06 GB), ≈ 0.9 GB of histogram-matching temporaries removed, and ingest of rasters under ≈ 700 Mpx no longer writes and re-reads a full-size intermediate.

5 · LIMITATIONS

- Accuracy figures are operator field validation, not a repository-held labelled benchmark; sample sizes and per-class breakdown are not recorded here (see §1 provenance note).
- One timing run per configuration on one machine; no variance characterised.
- **Ingest is excluded** from §3 — measuring the in-RAM COG staging means re-ingesting 24 GB twice.
- SAM refinement and learned land cover were disabled for the timing run, so 110.5 s is a lower bound on a full production analysis; including them lowers the overall speed ratio further.
- §3.1 per-operation figures are synthetic at 6144²; the IoU figures in §2 are from earlier recorded runs on different imagery and were not re-measured.

Sources: operator validation across multiple datasets (accuracy); A/B harness over rasters 16/17 on the production machine (performance); scipy / skimage / MapLibre equivalence checks (parity). Timing and parity figures reproducible from the benchmark scripts.